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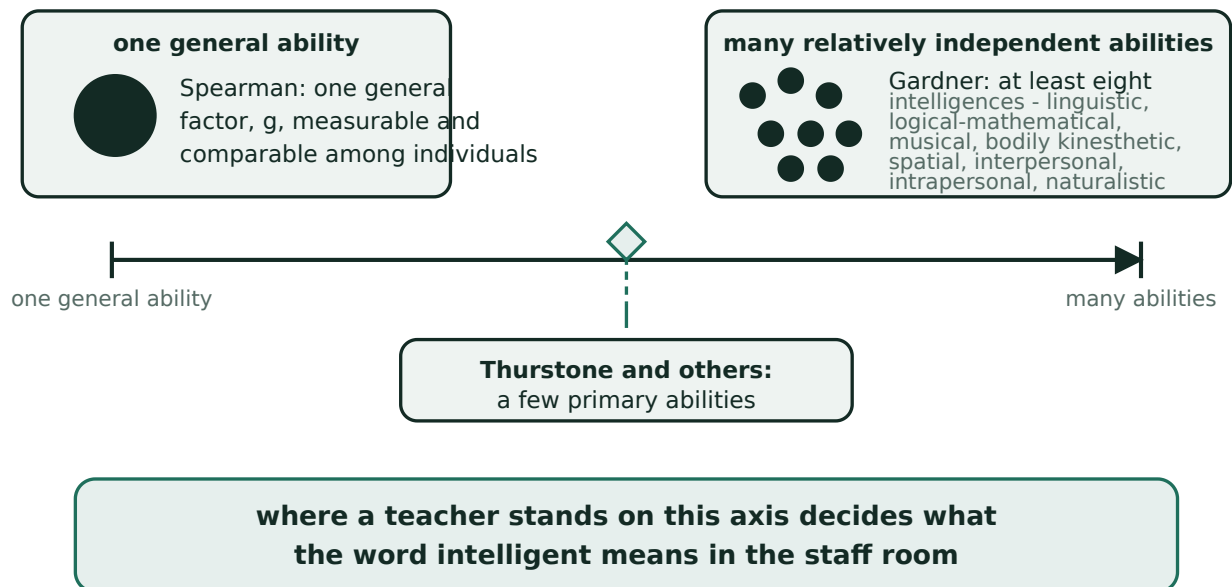
Topic 5.1 - Intelligence: theories and measurement

A real classroom situation

Two pupils sit in the same Class 6 in a Hyderabad school. Salman solves the mathematics textbook's hardest problems before the teacher finishes writing them, and cannot hold a conversation with the group beside him. Kiran struggles with every written test she has ever sat, and when the school's fan broke last June, it was Kiran who diagnosed the loose wire while three teachers watched.

The staff room calls Salman intelligent and Kiran "weak in studies". Both words are doing the work of a theory. Which theory - and what it entitles a school to conclude - is this topic's business.

Intelligence theories: one axis, different meanings



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Explanation

Theories of intelligence: one ability or many

What is intelligence? The answers arrange themselves along one axis (Spielman, Jenkins & Lovett, 2020).

At one end, **one general ability**. Charles Spearman, working from the observation that pupils who do well on one kind of test tend to do well on others, "believed intelligence consisted of one general factor, called g, which could be measured and compared among individuals" (Spielman, Jenkins & Lovett, 2020). On this view Salman's mathematics marks predict his conversation skills, and Kiran's test marks predict her hands - and the staff room's single word for both pupils is, strictly, Spearman's word.

At the other end, **many abilities**. Howard Gardner argued that "each person possesses at least eight intelligences" - linguistic, logical-mathematical, musical, bodily kinesthetic, spatial, interpersonal, intrapersonal, and naturalistic (Spielman, Jenkins & Lovett, 2020). On this view Salman's logical-mathematical strength and Kiran's practical-spatial diagnosis are different intelligences, and a

single score that ranks them against each other is measuring one slice of each and calling it the whole.

Between them stand the middle positions - Thurstone's few primary abilities, and the modern two-level accounts - which keep a general factor while admitting distinct strands. Most working psychologists stand somewhere here: there is something general, and there are strands.

For a teacher, the axis is not a quiz to pass but a position to notice. **Where you stand decides what your words mean.** A staff room on Spearman's end has one ladder, with every pupil on a rung; a staff room at Gardner's end has a profile, with every pupil strong somewhere. Neither end is free: the ladder misses Kirin's wire, and the profile, unchecked, can flatter every pupil into a strength no one can name.

The misconception to clear now. "The multiple-intelligences theory is proven science, and anyone who doubts it doubts the pupils." The honest position is more interesting. Among cognitive psychologists, Gardner's theory "has been heavily criticized for lacking empirical evidence" (Spielman, Jenkins & Lovett, 2020) - the eight intelligences do not show up as separable factors in test data. And yet educators use it, because it names abilities schools have historically ignored. Both halves are true at once, and a teacher who can hold both is better armed than one who has picked a side.

Multiple intelligences: the claim and its evidence

The claim, stated fairly: human abilities are wider than the linguistic and logical-mathematical skills schools examine, and a pupil weak on those two may be strong elsewhere. The classroom appeal is real, and Vosniadou (2001) makes the adjacent point carefully: Gardner has argued there are "many dimensions of human intelligence other than the logical and linguistic skills that are usually valued in most school environments... Schools must create the best environment for the development of children taking into consideration such individual differences."

The evidence caution, stated just as fairly: the eight intelligences have not demonstrated independence in measurement. What has survived scrutiny is

narrower and still useful - abilities are multiple enough that a single score is a poor summary of a pupil, and the school-valued two are not the only two that matter.

A third claim circulates beside these and must be separated from both: the **learning-styles** claim that pupils learn best when taught in their preferred style (visual, auditory, kinaesthetic). The evidence for matching teaching to a diagnosed style is weak, and the theory is not the same as multiple intelligences, whatever the staff-room shorthand says. What survives is humbler and free: pupils differ, and a teacher who presents a new idea in more than one mode - said, shown, done - reaches more of them than one who only says it.

Measurement of intelligence: what tests measure, and what they miss

The measurement story, briefly. "IQ stands for intelligence quotient and describes a score earned on a test designed to measure intelligence" (Spielman, Jenkins & Lovett, 2020). The tests are standardised: on most, "the average (or mean) IQ score is 100", and "one standard deviation is 15 points", with scores from 85 to 115 counting as the average band - about two-thirds of the population.

Three honest limits belong beside those facts.

A score summarises test performance, not a pupil. It samples the abilities the test samples, under the conditions the test sets. Kiran's wire-diagnosis is not in the sample.

The tests were built by people, and history shows it. Early IQ testing was used to rank immigrants and justify exclusion, in ways the data never supported. A teacher who knows this history treats every score as one source of evidence, never a verdict.

The measurement question is itself open. Researchers still debate "the degree to which any test can truly measure an individual's intelligence" (Spielman, Jenkins & Lovett, 2020). A score predicts school performance moderately well - that is what it was built for - and predicts almost nothing else a school exists to grow.

what a test does	the fact	what it does NOT show	the teacher's stance
the score a number	a number from a standardised test: mean 100, deviation 15	a summary of performance on the sampled tasks	one source of evidence, never a verdict
the band 85 to 115	85 to 115 covers about two-thirds of people	that those outside the band are worth less	bands describe, they do not decide
the prediction school only	predicts school performance moderately well	almost nothing else a school grows	use for support decisions, not labels
the history remember	early tests ranked people unjustly	that today's tests are free of that risk	know the history, hold the humility

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Activity: The talent show

In small groups, about 20 minutes.

A Class 5 talent show runs for one hour: singing, cricket skills, story-telling, mimicry, a pupil who can name every district of Sindh and its crop, a pupil who repairs the head teacher's torch.

1. For each act, name the ability it displays - using Gardner's names where they fit and plain words where they do not.
2. Now the hard part: which of these abilities does the school's examination system sample? What follows from your answer?
3. Finally: has the talent show demonstrated the theory of multiple intelligences, or has it demonstrated that pupils differ? Explain the difference, using this topic's evidence caution.

Check your understanding

1. State Spearman's claim in one sentence, and say what it predicts about Salman and Kiran. (*Understand*)
2. Name any five of Gardner's eight intelligences. (*Remember*)
3. State the evidence caution about multiple intelligences fairly, in two sentences, without dismissing its classroom appeal. (*Understand*)
4. What does an IQ-style score predict moderately well, and what does it not predict? (*Understand*)
5. A colleague says "she is a visual learner, so I teach her visually." Separate the learning-styles claim from what the evidence supports, in three sentences. (*Apply*)

Summary

Theories of intelligence arrange along an axis from one general factor (Spearman's *g*) to many relatively independent abilities (Gardner's eight), with most working psychologists between. Gardner's theory carries a real classroom appeal and a real evidence caution, and holding both is the analytic skill. The tests measure what they sample - mean 100, deviation 15, the 85-115 average band - predict school performance moderately and little else, and carry a history that argues for humility. What survives every caution is this: pupils differ in more ways than examinations sample, and a single score is a poor summary of a person.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can state Spearman's and Gardner's claims and place them on the axis.
- ☐ I can state the multiple-intelligences evidence caution fairly, with its appeal.
- ☐ I can say what an IQ-style score is, what it predicts, and what it does not.
- ☐ I can separate the learning-styles claim from what the evidence supports.

Try this at your practicum school

Ask the cooperating teacher to name the three pupils the examinations rate highest, and the three the teacher would most want beside them if the school bus broke down on a hot road. Compare the two lists, and note what the comparison shows about what the tests sample.

Summative task

Write **250-300 words** on the staff room's two words.

The staff room calls Salman "intelligent" and Kiran "weak in studies". Using this topic, analyse what each label assumes (which theory, implicitly), what evidence in the description fits and does not fit, and what each label leads the school to do and stop doing. Close with the label you would use instead, and defend it.

Mini-rubric

Criterion	What a strong answer shows
Theory identification	Each label is read for its implicit theory, not just contradicted.
Evidence discipline	The fit and misfit of each label uses the described facts only. (<i>Analyze</i>)
Consequence analysis	What each label leads the school to do and stop doing is named.
Plain, accurate writing	Terms used correctly; no padding; within the word count.

Further reading

- Spielman, R., Jenkins, W., & Lovett, M. (2020). *Psychology 2e* (OpenStax), Sections 7.4-7.5. CC BY-NC-SA 4.0. <https://openstax.org/books/psychology-2e/pages/7-4-what-are-intelligence-and-creativity>

- Vosniadou, S. (2001). *How children learn*. Educational Practices Series No. 7. International Academy of Education / UNESCO IBE. <https://eric.ed.gov/?id=ED455955>

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